CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

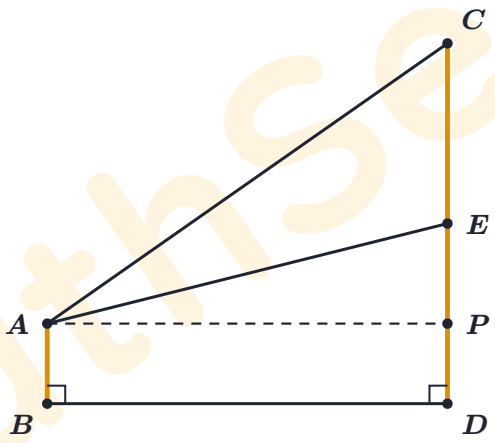
Chapter 22: Heights and Distances

Previous year board questions, topic-wise: 7 questions

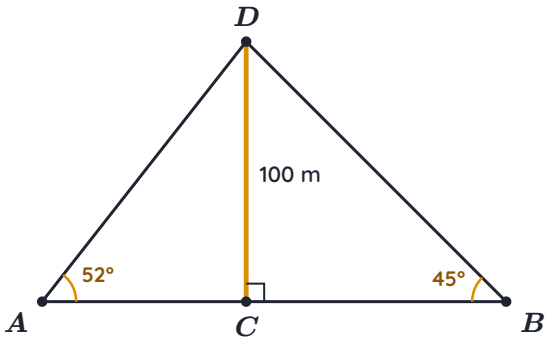
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Problems with two triangles

1. A vertical tower AB stands 100 m from the foot D of a building CD. E is the mid-point of CD. The angle of depression of the top A of the tower is 35° from C and 14° from E. Using tangent values rounded to two decimal places, find the height of (a) the tower (b) the building. ICSE 2025, 5 marks

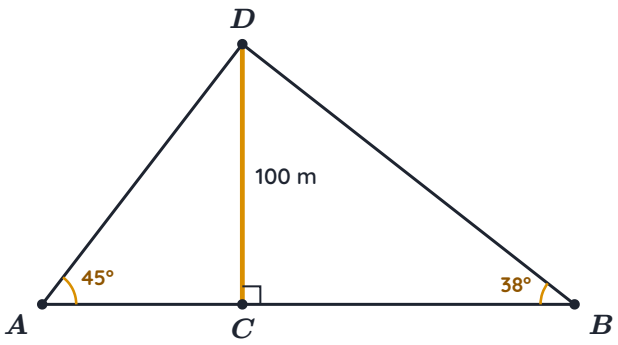


2. A tree CD is 100 m high. From two points A and B on the ground on opposite sides of the tree, its top D is seen at angles of elevation of 52° and 45°. Find the distance AB to the nearest metre.
ICSE 2024, 5 marks

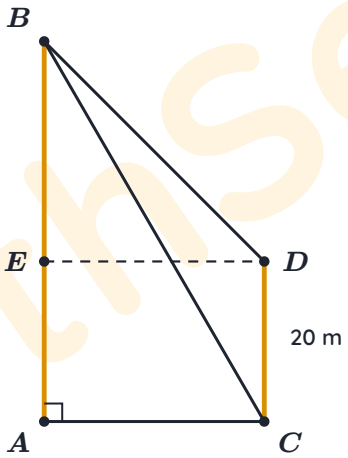


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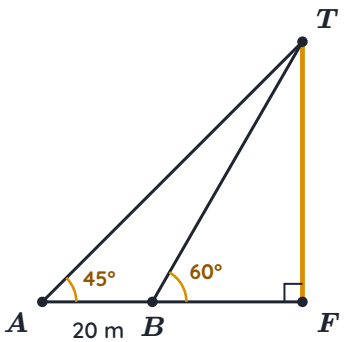
3. From the top D of a tower CD 100 m high, two ships A and B on opposite sides of the tower are seen at angles of depression of $45°$ and $38°$. The ships and the foot of the tower are in one straight line. Find the distance between the ships to the nearest metre. ICSE 2023, 4 marks



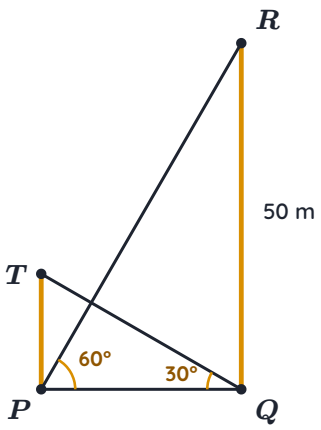
4. From the top of a cliff, the top and the bottom of a tower 20 m high are seen at angles of depression of $45°$ and $60°$. Find (a) the height of the cliff (b) the distance between the cliff and the tower. ICSE 2020, 4 marks



5. A man sees the top of a tower at an angle of elevation of $45°$. He walks 20 m straight towards the tower, and the angle of elevation becomes $60°$. Find the height of the tower correct to 2 significant figures. ICSE 2019, 4 marks



6. QR is a tower 50 m high and PT is another tower, with P and Q on level ground. From P the angle of elevation of R is $60°$, and from Q the angle of elevation of T is $30°$. Find the height of PT to the nearest metre. ICSE 2018, 4 marks



7. From the top of a lighthouse 60 m high, two ships on opposite sides of it are seen at angles of depression of $60°$ and $45°$. Find the distance between the ships to the nearest whole number.
ICSE 2017, 4 marks

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